NSF FINDERS FOUNDRY · AWARD NO. 2627515 · SUPPORTING DOCUMENT 2

# Glass-Box design specification

The lesson, design themes, AI guardrails, and standards alignment for a classroom AI that keeps its reasoning visible and the thinking with the student.

**VERSION** 1.0 **DATE** 2026 **INSTITUTION** UNIVERSITY OF LOUISIANA AT LAFAYETTE

## Overview

*Glass-Box* is a design for classroom AI that keeps the AI's reasoning visible and keeps the thinking with the student. This specification records the design at the end of the 2026 planning phase: the lesson it supports, the eleven themes that govern it, the AI's guardrails, and the standards the lesson addresses.

**The problem.** Most AI tools work as a black box: the answer appears, but the reasoning stays hidden.

**Three commitments.**

1. **Build thinking; don't replace it.** The AI questions, hints, and defines terms. It does not produce the student's work.
2. **Show how AI works.** Students learn how AI perceives, reasons, and learns while they use it, through live gauges, plain explanations, and stated limits.
3. **Balance screens with real experience.** Going outside, handling natural objects, journaling, building, and working with classmates take up most of the week. The AI provides important support along the way.

**Context.** Middle school science and engineering (grades 6 to 8). One week-long lesson in which students observe a living thing, learn how one of its abilities works, and invent something inspired by it. Two interfaces support the lesson: a student AI and a teacher view.

## The lesson: Biology to Design in five levels

Students move from noticing nature to presenting an invention over five levels, about one per class day. Biology comes first, design second, and the AI never takes over the work at any level. The model is Velcro: an engineer noticed burdock seeds clinging to his dog's fur, found the tiny hooks that made it happen, and turned that into a fastener.

| LEVEL | WHAT THE STUDENT DOES | WHAT THE AI DOES | OFF-SCREEN WORK |
|---|---|---|---|
| **1. Nature's abilities** | Goes outside (or uses brought-in natural objects or a virtual walk), observes a living thing with all their senses, then sketches and photographs it. | Says it can only see pixels and needs the student's senses. Explains how it recognized what is in the photo. Suggests abilities to look for. | Field journal entries; test one "I wonder" question at home. |

| LEVEL | WHAT THE STUDENT DOES | WHAT THE AI DOES | OFF-SCREEN WORK |
|---|---|---|---|
| **2. How it works** | Chooses one ability, guesses how it works, reads a short nature story or research excerpt, and explains the mechanism in their own words. | Defines any word but won't summarize the idea. Checks the explanation against the key steps and marks gaps as "not yet." Holds back existing inventions. | Notice everyday situations where people would want that ability. |
| **3. Connect to people** | Uses situation mapping ("When do people want to do this?") to pick a human need and form a first design idea. | Asks questions and suggests tests. Shows existing inventions only after the student has an idea. Tags which ideas came from whom. Alerts the teacher if a student is stuck. | Fill in the design table: inspiration, ability, mechanism, situation, idea. |
| **4. Design** | Sketches the design from several views with labels; may build a rough prototype; plans a presentation. | Checks that the labeled drawing explains the design and prompts for missing parts. | Mostly hands-on. |
| **5. Present** | Presents the invention to the class: the need, the inspiration, how it works, and open questions. | None. | Entirely off-screen. |

Each level is broken into chapters of about 15 minutes. A paper field journal carries the student's work between levels, and a level sticker marks each completed level.

## Eleven design themes

The co-design team consolidated its sessions, reviewer feedback, and user story comments into eleven themes (v3.1, September 2026). Each theme was tested with parents and teachers through one of five focus group modules.

| # | DESIGN THEME | WHAT IT MEANS IN THE DESIGN | MODULE |
|---|---|---|---|
| 1 | **Glass box: the AI shows its thinking on the live task, through instruments, not engine** | Live gauges (confidence, sources, reasoning depth) and an "X-ray" view show what the AI is doing on this task. Every gauge must map to something the system actually computes. | 1 |
| 2 | **A machine, not a friend, but one that genuinely listens** | A robot persona states its limits when they matter ("I can only see pixels") and restates what the student actually said, with no empty praise. | 2 |
| 3 | **The intelligence is the student's, and the shared load stays visible** | The AI asks, points to readings, and defines terms but never authors deliverables. Tags show which ideas came from the student and which from the AI. | 3 |
| 4 | **The body does the learning** | Students go outside (or use brought-in objects or virtual walks), observe with senses the camera lacks, and sketch before uploading. | 3 |
| 5 | **The companion journal carries the structure** | A paper field journal holds noticings, sketches, and design tables before the AI sees them, scaffolds each level, and links class sessions through homework. | 3 |

| # | DESIGN THEME | WHAT IT MEANS IN THE DESIGN | MODULE |
|---|---|---|---|
| 6 | **Conversational pacing: short turns, many of them, every question answered** | Two to three sentences per AI turn, a cap on AI turns before the student acts, no rhetorical questions, and text rendered at reading speed. | 4 |
| 7 | **Structure and agency: chapters, levels, and a map you can move around on** | About 15-minute chapters, light leveling (curiosity, creativity, collaboration), and a design-cycle map students can move back through to revise. | 4 |
| 8 | **Teachers keep professional judgment, and get their own instruments** | A teacher view shows where students are and who is stuck, with chat summaries and suggested prompts, supporting teachers rather than replacing them. | 5 |
| 9 | **Accessible by design** | A biology-first entry point, grade-level curated readings in place of open web search, and no timed tasks that favor fast typers. | 4 |
| 10 | **Responsive instruction and assessment built in** | Targeted follow-up questions in place of more reading, plans for off-track replies, and formative assessment defined at each level. | 5 |
| 11 | **AI literacy and workforce content as a pipeline** | Explainers on how neural networks and generative AI work, and videos of people who build AI systems, linking the lesson to future careers. | 1 |

Modules: (1) *Glass-Box* instrumentation, (2) AI limits and robot persona, (3) companion journal integration, (4) Socratic pacing and navigation, (5) teacher view and steering.

## Guardrails: what the AI does and won't do

Each guardrail has two halves, and both are binding: what the AI does, and what it won't do in its place. Dropping the second half turns a constraint into a suggestion.

### Student AI

| THE AI DOES | THE AI WON'T |
|---|---|
| Ask questions that hand the thinking back to the student | Write, finish, or rewrite the student's work |
| Define any word, as often as the student asks | Summarize a mechanism the student is meant to explain |
| State what it can't sense at the moment it matters ("I can only see pixels") | Claim senses, feelings, or a human identity |
| Show its confidence and sources, and explain on request how it reached a response | Present answers without showing what they rest on |
| Show existing inventions once the student has an idea of their own | Show existing inventions first, inviting students to copy them |
| Restate what the student said, in turns of two or three sentences | Offer empty praise, ask rhetorical questions, or lecture |
| Make the step smaller when a student is stuck (a hint, then a sentence starter) | Give in and supply the answer |
| Point to curated, grade-level readings | Send students to open web search |

### Teacher view

| THE TEACHER VIEW DOES | THE TEACHER VIEW WON'T |
|---|---|
| Show where each student is, by level and chapter | Make instructional decisions for the teacher |
| Alert the teacher when a student keeps asking for answers, with what the AI already tried | Hide what happened between the student and the AI |
| Suggest ways the teacher could step in | Replace the teacher's conversation with students |

**Open design decisions** to settle in the next phase:

- A firm limit on AI turn length.
- How to tell warmth apart from cognitive offloading when evaluating transcripts.
- Whether the student or the AI makes the generalization step in the coaching arc.
- Whether to keep existing inventions hidden until the student has an idea, and how the AI should explain that choice.

## Standards alignment

The Biology to Design lesson addresses the NGSS middle school engineering design standards (MS-ETS1-1, MS-ETS1-2, MS-ETS1-3) and two NGSS crosscutting concepts, with Common Core literacy connections. The AI literacy strand connects to the AI4K12 Five Big Ideas.

| STANDARD | HOW THE LESSON ADDRESSES IT | AI LITERACY ALIGNMENT (AI4K12 GRADES 6-8; AILIT) |
|---|---|---|
| **NGSS Practice:** Asking Questions and Defining Problems (MS-ETS1-1) | Students define a design problem while mapping how a nature-inspired idea could meet a human need or want. They develop an object, tool, process, or system that meets multiple criteria and constraints, using scientific knowledge to shape possible solutions. | AI4K12 5-A-i: students compare how human needs shape their designs with how stakeholders' goals shape AI systems (Level 5 reflection). AILit Engage with AI: students weigh the AI's questions and suggestions while defining the problem. |
| **NGSS Core Idea ETS1.A:** Defining and Delimiting Engineering Problems (MS-ETS1-1) | Criteria and constraints shape each design; one required criterion is that the solution is bioinspired. Students identify constraints, including where more work would be needed with more time, and use scientific knowledge to drive ideation. | AI4K12 1-A-i, 1-A-ii: students contrast their own multisensory observation with the AI's camera-only sensing, a limit on what the AI can contribute (Level 1). AILit Manage AI: students decide which parts of the task the AI is suited for. |
| **NGSS Core Idea ETS1.B:** Developing Possible Solutions (MS-ETS1-2, MS-ETS1-3) | Students follow a routine process to evaluate their design concept against criteria (bioinspiration, meeting human needs or wants) and constraints (scientific understanding of biological function). | AI4K12 2-A-iv: word embeddings explain how the AI matches a nature story to a student's idea; 3-C-iii: training data shapes what the AI suggests (Levels 2 and 3). AILit Engage with AI: students check AI suggestions against the design criteria. |

| STANDARD | HOW THE LESSON ADDRESSES IT | AI LITERACY ALIGNMENT (AI4K12 GRADES 6-8; AILIT) |
|---|---|---|
| **NGSS Crosscutting Concept:** Influence of Science, Engineering, and Technology on Society and the Natural World (MS-ETS1-1) | Designs respond to human needs and wants and draw on the findings of scientific research. | AI4K12 5-B-i, 5-C-ii: students consider how AI is changing daily life and work, including profiles of people who build AI (Level 5). AILit Shape AI: people decide how AI is built and used. |
| **NGSS Crosscutting Concept:** Systems and System Models | Students explain a living thing's ability as a system of interacting parts (Level 2), then design their invention as a system that meets criteria and constraints (Levels 3 and 4). | AI4K12 1-B-iv, 3-B-i: students see the AI itself as a system of interacting parts, from feature detectors to a neural network (computer vision explainer; X-ray view). AILit Shape AI: understanding how AI systems work. |
| **Common Core ELA/Literacy:** RST.6-8.1, RST.6-8.7, RST.6-8.9, WHST.6-8.7, WHST.6-8.8, WHST.6-8.9, SL.8.5 | Students read and cite nature stories and research excerpts (Level 2), connect text with sketches and diagrams (Levels 1 and 4), and present their design with visuals (Level 5). | AI4K12 1-B-iii: how a computer recognizes phrases from sequences of words (language model explainer). AILit Engage with AI: students check the sources behind an AI reply with the "What I'm using" gauge (Level 2). |

*AI4K12 codes refer to the grade 6-8 learning objectives in the AI4K12 Five Big Ideas progression charts. AILit domains (Engage with AI, Create with AI, Manage AI, Shape AI) are from the AI Literacy Framework of the European Commission and the OECD.*

## About this document

**How it was developed.** An interdisciplinary team co-designed *Glass-Box* in weekly Agile sprints, with K-12 teachers and a parent advocate as co-designers, not research subjects. Parents and teachers reviewed the design in two live focus groups and through an asynchronous option, under UL Lafayette IRB approval IRB-26-1004-CILAT. Version 1.0 reflects the design before those findings are incorporated.

**See it in action.** The project site at glassboxlearningai.github.io hosts the interactive *A Week with Maya* walkthrough, eight narrated videos, and the clickable prototypes. Supporting document 1 shows screens from each.

**Team.** Douglas C. Williams (PI), Manyu Li (co-PI), and Chee-Hung Chu (co-PI), University of Louisiana at Lafayette; Sam Stier, LearningWithNature.org; Robert Gilson, Dwight School; Carolyn Haruo, John F. Kennedy High School, Guam; Priya Cook, parent advocate.

**Acknowledgment.** This material is based upon work supported by the National Science Foundation under Grant No. 2627515. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.